

SYLTRA

Smart Living. Seamlessly Connected.

Integrated Management System Blueprint

ISO 9001:2015 · ISO 14001:2015 · ISO 45001:2018

Quality, Environmental & Occupational Health and Safety

Site safety protocols, audit governance, and vendor alliance strategy

1. Integrated Management System (IMS) Framework

SYLTRA operates a single Integrated Management System (IMS) that fuses Quality (ISO 9001:2015), Environmental (ISO 14001:2015), and Occupational Health & Safety (ISO 45001:2018) management into one unified operating model, purpose-built for automation, smart-systems integration, and field maintenance services. Rather than running three parallel systems, SYLTRA aligns policy, risk assessment, document control, audit cycles, and management review into a shared governance structure — reducing duplication while raising the consistency of evidence collected across every engagement.

The IMS follows the Plan-Do-Check-Act (PDCA) cycle at both the corporate and project level: strategic objectives and risk registers are planned annually, execution is carried out under controlled procedures and toolbox talks, performance is checked through internal audits and KPI tracking, and corrective actions feed back into policy and training updates.

Standard	Core Focus for SYLTRA	Key Output Records
ISO 9001:2015	Accuracy in PLC/SCADA programming, precision wiring, hardware verification, and zero-defect commissioning across smart-home and industrial automation deployments.	FAT/SAT records, Customer Satisfaction Surveys, Non-Conformity Reports (NCR).
ISO 14001:2015	Safe collection, categorization, and ecological disposal of industrial electronic waste, spent batteries, and wiring scrap generated during installation and retrofit projects.	E-Waste Log, Disposal Certificates, Environmental Aspect Register.
ISO 45001:2018	Mitigation of live electrical shock hazards, lock-out tag-out (LOTO) protocols, working-at-height controls, and safe installation practices on cable trays and control panels.	Electrical Work Permits, Toolbox Talks Ledger, Incident/Near-Miss Register.

1.1 Integration Governance

A single IMS Representative (or the appointed Quality & HSE Manager) owns cross-standard coordination, ensuring that a change raised under one standard — for example a new hazardous-waste handling requirement — is automatically screened for impact on the quality and safety procedures it touches. Document control, internal audit scheduling, and management review are run as shared processes rather than three separate calendars.

2. Site Safety & Commissioning Protocols (Toolbox Talks)

Every operational morning, site managers conduct a short briefing (Toolbox Talk) with technical personnel before execution begins. Attendance is logged in the Toolbox Talks Ledger and cross-referenced against the day's work permits.

Protocol A — Live Electrical Panel Safety & LOTO

- **Mandatory Check:** Always use calibrated multimeters to confirm voltage absence before touching contacts.
- **Isolation:** Execute Lock-Out Tag-Out (LOTO) on source circuit breakers; only the active technician holds the key.
- **PPE:** Class 00 / Class 0 insulated gloves and insulated safety footwear are strictly mandatory.
- **Verification:** A second technician independently re-checks isolation before work commences on any panel rated above 230V.

Protocol B — Safe Programming & On-Site Testing (PLC/SCADA)

- **Backup Verification:** Never push modified logic or code without saving a local, time-stamped restore point.
- **Physical Clearance:** Before switching a PLC from Stop/Safe mode to Run mode, confirm all mechanical areas are clear of personnel.
- **Dual-Loop Verification:** Signal loop checks are synchronized between the software desk and the field technician via wireless communication.
- **Rollback Plan:** Every commissioning session defines a documented rollback path in case new logic causes unexpected behavior.

Protocol C — Working at Height & Cable Tray Installation

- **Fall Protection:** Full-body harness with a certified anchor point is mandatory above 1.8m, per approved scaffolding or lift equipment only.
- **Load Rating:** Cable trays and supports are inspected for rated load capacity before routing new cable runs.
- **Housekeeping:** Tools and materials are tethered or bagged to eliminate drop hazards to personnel below.

Protocol D — Emergency Response on Site

- Class CO2 fire extinguishers rated for electrical fires are staged within 10 meters of every active panel work area.
- 100% of field technicians are trained in electrical-shock rescue and basic first aid, refreshed annually.
- The main disconnect (Emergency Switch) is identified and communicated to all personnel at the start of every shift.

3. Annual Internal Audit Schedule

To guarantee external audit readiness, SYLTRA executes a cross-departmental, impartial internal audit scheme. Auditors are rotated so that no individual audits their own department, and findings are logged against the Non-Conformity & Corrective Action procedure with defined closure deadlines.

Department	Targeted Clauses (IMS)	Audit Focus Points
Engineering & Design	ISO 9001: Clause 8.3	Software revision control, backup logs, design-input validation, and change-management sign-off.
Site Operations	ISO 45001 / ISO 14001	Execution of LOTO logs, hazardous e-waste containment, PPE adherence, and toolbox talk completion rates.
Procurement / Stores	ISO 9001: Clause 8.4	Vendor evaluations, calibration validity of testing equipment, and traceability of critical components (PLCs, sensors).
Customer Support & Maintenance	ISO 9001: Clause 9.1	SLA adherence, response-time logs, and closure quality of maintenance tickets.
HR & Training	ISO 9001 / ISO 45001	Competency records, safety certification validity, and onboarding checklist completion for field staff.

3.1 Audit Methodology

Each audit cycle runs annually at minimum, with high-risk departments (Site Operations, Procurement) audited semi-annually. Auditors use pre-approved checklists, collect objective evidence (photographs, signed records, system logs), and issue findings within five working days. Non-conformities are classified as Major, Minor, or Observation, each carrying a defined maximum closure window (30 / 60 / 90 days respectively).

4. Management Review Agenda

An annual executive meeting verifies the structural health and compliance of SYLTRA's operational framework. High-risk periods (post-major-incident, or after a significant scope change) trigger an additional interim review.

Input Parameters

- Status of actions from previous management reviews.
- Changes in external and internal issues relevant to the IMS.
- Performance and effectiveness of the IMS, including trends in non-conformities, audit results, and customer satisfaction.
- Adequacy of resources (people, equipment, budget) to sustain operations.
- Effectiveness of actions taken to address risks and opportunities.
- Opportunities for continual improvement.

Decision Outputs

- Resource allocations for the following operating period.
- Optimization or modification of infrastructure, tooling, and software platforms.
- Targeted training pipelines and certification plans for engineering teams.
- Any need for changes to the IMS, including policy or objectives.

5. Global Vendor Alliance Strategy (Siemens / Schneider Electric)

The strategic path to achieve official System Integrator status and accelerate commercial acquisition velocity with tier-one automation vendors:

- 1. Talent Pipeline Certification:** Enroll a minimum of two engineers in official certification pipelines (Siemens TIA Portal ST-PRO or Schneider EcoStruxure Expert) within the current fiscal year.
- 2. Track Record Compilation:** Document three successful automation case studies showcasing proper wiring, programming standards, and clean electrical panel assembly, each backed by FAT/SAT evidence.
- 3. Compliance Sync:** Leverage this ISO-based IMS documentation framework to satisfy the rigorous business-management requirements demanded during vendor partner audits.
- 4. Formal Application & Technical Review:** Submit the System Integrator application package with certification evidence, case studies, and IMS audit summary; host the vendor's technical review visit.
- 5. Ongoing Partnership Maintenance:** Maintain certification currency, submit annual project volume reports, and re-validate engineer certifications on the vendor's renewal cycle.

5.1 Timeline Overview

Phase	Target Window	Owner
Certification enrollment	Month 1–2	Engineering Manager
Case study documentation	Month 2–4	Project Leads
Application submission	Month 5	Quality & HSE Manager
Technical review visit	Month 6	Executive Sponsor
Certification renewal cycle	Annual	Engineering Manager

6. Roles & Responsibilities (RACI Summary)

The table below summarizes accountability across the core IMS processes: **R**=Responsible, **A**=Accountable, **C**=Consulted, **I**=Informed.

Process	Executive Mgmt	Quality & HSE Mgr	Site/Project Lead	Technicians
Document Control	I	A/R	C	I
Internal Audit	I	A/R	C	I
Toolbox Talks	I	C	A/R	R
NCR & Corrective Action	I	A	R	C
Management Review	A/R	R	I	I
Vendor Evaluation	I	C	A/R	—

7. Document Control & Revision History

Version	Date	Description of Change	Approved By
1.0	Initial issue	First integrated blueprint combining ISO 9001/14001/45001 scope.	Quality & HSE Manager
2.0	Current issue	Rebranded to SYLTRA; expanded safety protocols, audit scope, RACI, and vendor timeline.	Quality & HSE Manager

Disclaimer: This system documentation blueprint is compiled for corporate structural development purposes based on ISO guidance and standard engineering practices. All field testing and execution must strictly respect active country laws and engineering safety declarations.